

External Loads:

$$P, F, A_x, A_y$$

Internal Loads: (N, kN, lb, kip):

N – Normal Load

V – Shear Load

$$\text{Double Shear: } V_A = \frac{F_A}{2}$$

Stress (GPa, MPa, psi, ksi):

Normal Stress: σ (sigma)

Shear Stress: τ (tau)

$$\sigma = \text{Stress} = \frac{\text{Force}}{\text{Area}}$$

$$\sigma_{\text{avg}} = \frac{N}{A}$$

$$\tau_{\text{avg}} = \frac{V}{A}$$

$$F.S. = \frac{\sigma_{\text{fail}}}{\sigma_{\text{allow}}} \text{ or } \frac{\tau_{\text{fail}}}{\tau_{\text{allow}}}$$

Average Normal Stress = Direct Average Shear Stress =

Direct Shear Stress = Simple Shear Stress

$$\text{Bearing Stress (Base Plate): } A_{\text{req'd}} = \frac{P}{(\sigma_b)_{\text{allow}}}$$

$$\text{Direct Shear Stress (Rod): } l_{\text{req'd}} = \frac{P}{\tau_{\text{allow}} \pi d}$$

Internal Stress Convention:

Tension Positive $\leftarrow (+)$. _

Compression Negative $\rightarrow (-)$ < -

Strain:

$$\text{Normal Strain: } \epsilon_{\text{avg}} = \frac{L_{\text{final}} - L_{\text{initial}}}{L_{\text{initial}}} = \frac{\delta}{L_0}$$

$$\text{Shear Strain: } \frac{\pi}{2} - \theta_{\text{rad}}$$

$$\text{Positive Shear Strain: } \theta < \frac{\pi}{2}$$

$$\text{Negative Shear Strain: } \theta > \frac{\pi}{2}$$

Mechanical Properties:

Hooke's Law: $\sigma = E$

$$\text{Mod. of Elasticity: } \epsilon = \frac{\sigma_2 - \sigma_1}{E_2 - E_1}$$

Elastic Deformation of an Axially Loaded Member:

$$\text{Single Load/Segment: } \delta = \frac{NL}{AE}$$

$$\text{Multiple Loads/Segments: } \Sigma \frac{NL}{AE} = \frac{N_1 L_1}{A_1 E} + \frac{N_2 L_2}{A_2 E} + \dots$$

Sign Convention: Both N and delta are positive if they cause tension and elongation:

$$[---] \rightarrow (+) \quad (+) \leftarrow [---]$$

Elastic Recovery:

$$E \text{ (Stiffness): } \frac{\sigma_y}{\epsilon_y \text{ (elastic recovery)}}$$

$$\epsilon_{\text{permanent}} = \epsilon_{\text{original segment}} - \epsilon_{\text{elastic recovery}}$$

$$L_{\text{final}} = L + L \epsilon_{\text{permanent}}$$

Statically Indeterminate:

(A) Force Equilibrium

(B) Compatibility Condition

(C) Use principle of superposition

Thermal Stress w/o Spring:

Gap = Thermal Displacement + Mechanical Displacement

$$\text{Gap} = \delta_{\text{Thermal}} + \delta_{\text{Mechanical}}$$

$$\text{(One Material) Gap: } = \delta_T + \delta_N$$

$$\begin{aligned} \text{(Two Material) Gap: } &= (\delta_{1-T} + \delta_{2-T}) + (\delta_{1-N} + \delta_{2-N}) \\ &= (\delta_{1-T} + \delta_{1-N}) + (\delta_{2-T} + \delta_{2-N}) \end{aligned}$$

$$\delta_T = \alpha \Delta T L$$

$$\delta_N = \frac{NL}{AE}$$

Thermal Stress w/ Spring:

$$F = kx$$

Stress Concentration:

$$\sigma_{\text{avg}} = \frac{N}{A_{\text{smallest}}}$$

$$K = \frac{\sigma_{\text{max}}}{\sigma_{\text{avg}}}$$

$$\sigma_{\text{allow}} = \sigma_{\text{max}} = K \sigma_{\text{avg}} = K \frac{N}{A}$$

Elastic Recovery:

Rigid Pipe w/ Guy Wire (Find Final Length)

If length, find strain!

- 1.) FBD
- 2.) EoE, find F
- 3.) Calculate σ of target using F
- 4.) Find corresponding ϵ from graph wro σ using ratio
- 5.) Find $\epsilon_{\text{elastic recovery}}$
- 6.) Find $\epsilon_{\text{permanent}}$
- 7.) Find $L_{\text{final}} = L + L\epsilon_{\text{permanent}}$ (ans)

Rod w/ Supports (Find Stress in Each Segment, Displacement)

- 1.) FBD
- 2.) Cut between letters
- 3.) Find σ of each segment (include T or C) (ans)
- 4.) Find EA
- 5.) Find $\Sigma \frac{NL}{AE}$ (ans)

Statically Indeterminate:

1. Force Equilibrium
2. Compatibility Condition

$$\delta_{AB} = 0$$

$$\delta_{AC} + \delta_{CB} = 0$$

$$\frac{N_{AC} + L_{AC}}{AE} + \frac{N_{CB}L_{CB}}{AE} = 0$$

3. Link internal force (N) to an external applied force (F) through the FBD analysis.
Given: AE, L_AC, N_CB = F_B, L_CB.

Thermal Stress w/o Spring:

What larger T2 is required to just close the gap?

$$\delta_T = \alpha_{al}\Delta TL_{al} + \alpha_{cu}\Delta TL_{cu}$$

Solve for T2 in ΔT

Determine average normal stress in rod if T2 = T?

$$\alpha_{al}\Delta TL_{al} + \alpha_{cu}\Delta TL_{cu} - 0.01 = -\left(\frac{NL_{Al}}{A_{Al}E_{Al}} + \frac{NL_{Cu}}{A_{Cu}E_{Cu}}\right)$$

$$\sigma_{Al} = \sigma_{Cu} = \frac{N}{A} \text{ (Ans)}$$

Thermal Stress w/ Spring

- 1.) Compatibility Condition

$$\begin{aligned}\delta_{Net} &= \delta_T + \delta_{\text{Spring(Reaction)}} \\ &= \alpha\Delta TL + \frac{-N_{\text{Spring(Reaction)}}L_{\text{rod}}}{A_{\text{rod}}E_{\text{rod}}}\end{aligned}$$

$$F = kx = k * (\delta_{Net} + x \text{ compressed}) \text{ (ans)}$$

Stress Concentration

1. Find shoulder K value
 - a. Find shoulder ratio r/h and w/h
 - b. Use lowest ratio to find K according to given graph
 - c. Find N from σ_{max} equation 3
2. Find K hole K value
 - a. Find hole ratio 2r/w = diameter/w
 - b. Use ratio to find K according to given graph
 - c. Find N from σ_{max} equation 3
3. Lowest N value is allowable max force

